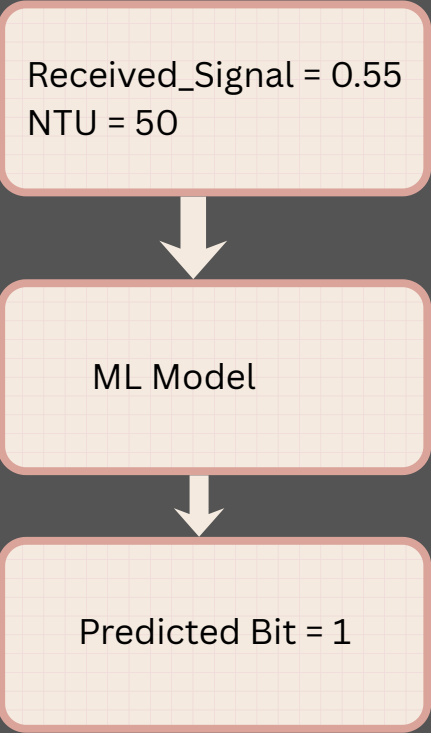


# Day 1 - Problem Understanding & System Definition

## Problem Statement :

- Information is transmitted as binary bits (0 and 1) using optical signals underwater.
- Turbidity (NTU) causes absorption and scattering of light in water.
- This changes the received optical signal amplitude.
- As a result, fixed-threshold detection becomes unreliable and may cause bit errors.
- The dataset contains received analog signals, NTU values, and transmitted bit labels (0/1).
- The goal is to develop supervised ML models that accurately recover transmitted bits under different turbidity conditions.

| Received_Sign | NTU | Transmitted_B |
|---------------|-----|---------------|
| 0.12          | 5   | 0             |
| 0.18          | 5   | 0             |
| 0.85          | 5   | 1             |
| 0.92          | 5   | 1             |
| 0.25          | 20  | 0             |
| 0.35          | 20  | 0             |
| 0.72          | 20  | 1             |
| 0.81          | 20  | 1             |
| 0.4           | 50  | 0             |
| 0.55          | 50  | 1             |
| 0.63          | 50  | 1             |



## Pipeline | Workflow :

